

Basic Principles of Medicine 1
Module: Musculoskeletal System
Lecture #: 21

Calcium, Phosphate & Vitamin D in Bone Health

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Recommended Reading

- **Lippincott's Biochemistry:**

<https://premiumbasicsciences.lwwhealthlibrary.com/content.aspx?sectionid=260214405&bookid=3402>

- Chapter 28: **Section: Vitamin D:** Pages 437 – 440 [Micronutrients: Vitamins | Lippincott Illustrated Reviews: Biochemistry, 9e | Premium Basic Sciences | Health Library](#)
- Chapter 29: **Section: Macrominerals: Calcium and phosphorus:** Pages 446 – 447 [Micronutrients: Minerals | Lippincott Illustrated Reviews: Biochemistry, 9e | Premium Basic Sciences | Health Library](#)
- (Summary: page 442-443 and 455)
- Study Questions: 28.7 and 29.6

Required pre-reading: DLA: Vitamin D

SOM.MK.III.BPM1.2.MSK.1.BCHM.0200	Describe vitamin D formation in the skin and its conversion to calcitriol (in liver and kidney) indicating factors affecting this process.
SOM.MK.III.BPM1.2.MSK.1.BCHM.0201	Describe the functions of 1,25 dihydroxy-D (calcitriol) related to calcium and phosphate metabolism
SOM.MK.III.BPM1.2.MSK.1.BCHM.0202	Indicate vitamins that act as steroid hormones (calcitriol)
SOM.MK.III.BPM1.2.MSK.1.BCHM.0203	Indicate clinical features and manifestations of hypervitaminosis D
SOM.MK.III.BPM1.2.MSK.1.BCHM.0204	Describe the basis of clinical manifestation in rickets and osteomalacia. Identify risk factors for development of these disorders.

Learning objectives: Calcium, Phosphate and Vitamin D in Bone Health

SOM.MK.I.BPM1.2.MSK.1.BCHM.0205	Indicate the nutritional needs as it relates to calcium, phosphate and vitamin D
SOM.MK.I.BPM1.2.MSK.1.BCHM.0206	Describe the role of calcium, phosphate and vitamin D in nutrition and in the development of bone disease
SOM.MK.I.BPM1.2.MSK.1.BCHM.0207	Describe the various functions of calcium and phosphate in humans
SOM.MK.I.BPM1.2.MSK.1.BCHM.0208	Identify the three forms of extracellular calcium, and their contribution to total extracellular calcium
SOM.MK.I.BPM1.2.MSK.1.BCHM.0209	Describe hypocalcemia and hypercalcemia as well as their causes, manifestations, diagnosis and treatment
SOM.MK.I.BPM1.2.MSK.1.BCHM.0211	Describe the associations among intestinal absorption, renal elimination, bone stores, and the function of calcitriol and parathyroid hormone in regulating serum calcium and phosphate
SOM.MK.I.BPM1.2.MSK.1.BCHM.0212	Correlate the skeletal and non-skeletal actions of vitamin D; reasons for and likely extent of its deficiency; likely deficiency diseases and potential solutions in terms of food fortification and supplementation.
SOM.MK.I.BPM1.2.MSK.1.BCHM.0213	Describe lab findings (elevated alkaline phosphatase) in osteomalacia and rickets.
SOM.MK.I.BPM1.2.MSK.1.BCHM.0214	Correlate hereditary resistance to vitamin D (vitamin D resistant rickets) to the biochemical mechanisms

Calcium



- Serum calcium:
 - **Bound** to plasma proteins (notably **albumin**), 40 – 45%
 - Bound to anions (e.g., phosphate, citrate), 15%
 - **Free or ionized state, 40 – 45%, (Metabolically active form)**
- Total serum calcium range: 8.5-10.5 mg/dL (2.12 – 2.62 mmol/L)
- Dietary sources:
 - Milk, cheese, dairy products
 - Green leafy vegetables
 - Soya beans, nuts, fish
- Dietary requirements:
 - RDA: 25-30 mmol (1000-1200 mg) for adults
 - Long-term low calcium intake impairs peak bone mass development
 - Requirements increases during pregnancy, lactation and childhood

Functions of Calcium

- Provides mechanical stability in bone
 - 99% of total calcium present in skeleton
 - Provides mechanical support
 - Serves as a reservoir of calcium and phosphorus
- Stabilizes resting membrane potential of excitable cells
 - Hypocalcemia → increases membrane excitability
 - Hypercalcemia → decreases membrane excitability
- Second messenger in signaling pathways (diacylglycerol/ IP_3)
- Cofactor for enzymes (e.g., phospholipase A_2 , blood clotting)
- Regulates muscle contraction



Absorption and Excretion of Calcium is Highly Regulated

PTH = Parathyroid Hormone

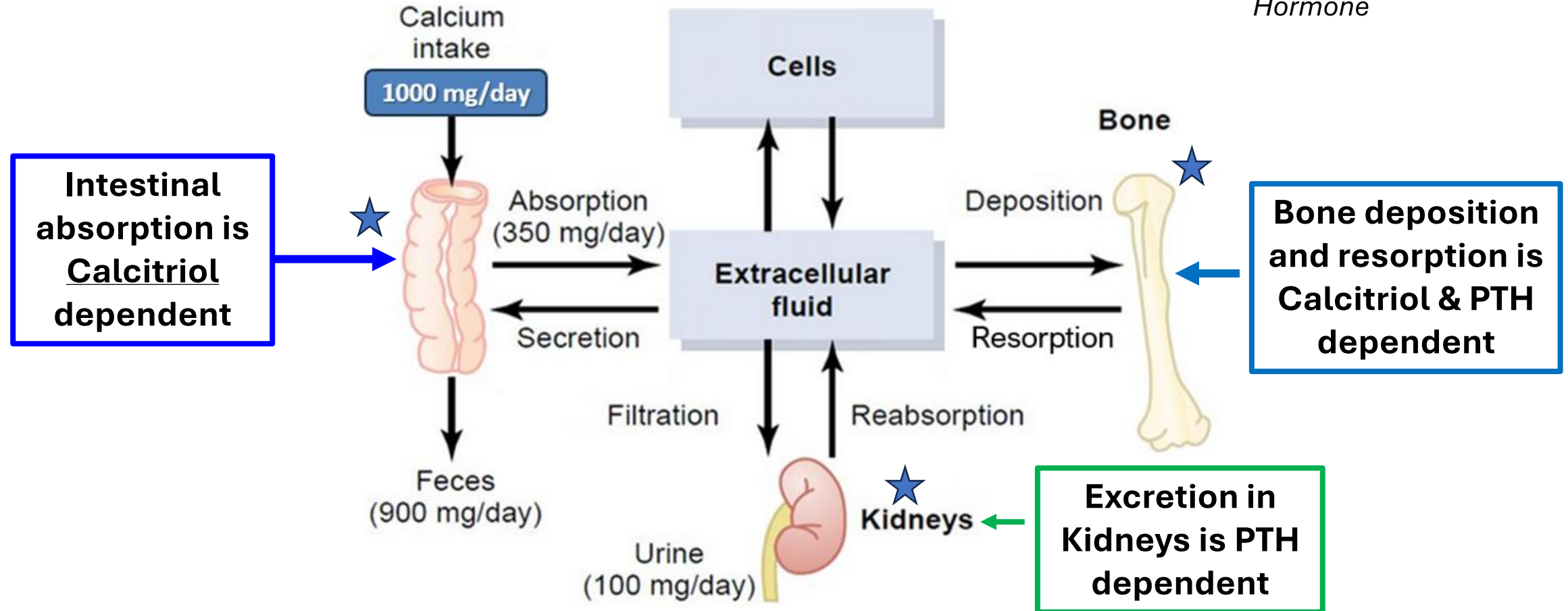
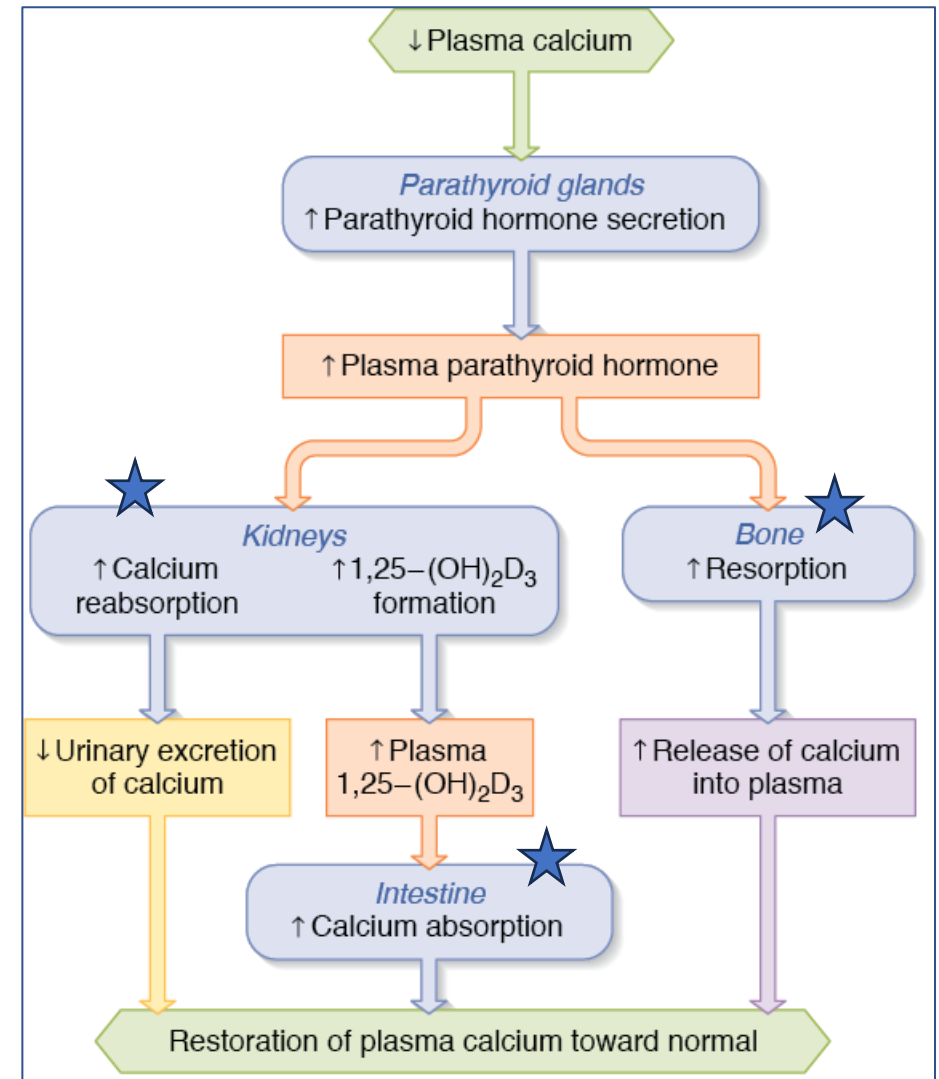


Image adapted from: Hall, J. E. (2015). Guyton and hall textbook of medical physiology (13th ed.). W B Saunders.

Calcitriol is active form of Vitamin D (aka 1,25-dihydroxycholecalciferol or $1,25(\text{OH})_2\text{D}_3$ or $1,25(\text{OH})_2\text{D}$)

Hormonal Regulation of Calcium

- Serum calcium levels regulated by
 - **Parathyroid hormone** (PTH) and
 - **Vitamin D**
 - Calcitonin
- Hormonal effects on ★
 - Bone
 - Kidney, and the
 - Gastrointestinal tract (Intestine)



Phosphorus

- 85% of total body phosphorus is found in bone mineral
- Forms of phosphorous (ECF and ICF): H_2PO_4^- , NaHPO_4^- , and HPO_4^{2-}
- Efficiently absorbed in **intestine**
 - Absorption enhanced by 1,25-dihydroxyvitamin D (active vitamin D)
- Urinary excretion of phosphate depends on adequate filtration of phosphate in the glomerulus



Regulation of Phosphorus

- **Phosphorus levels regulated by:**

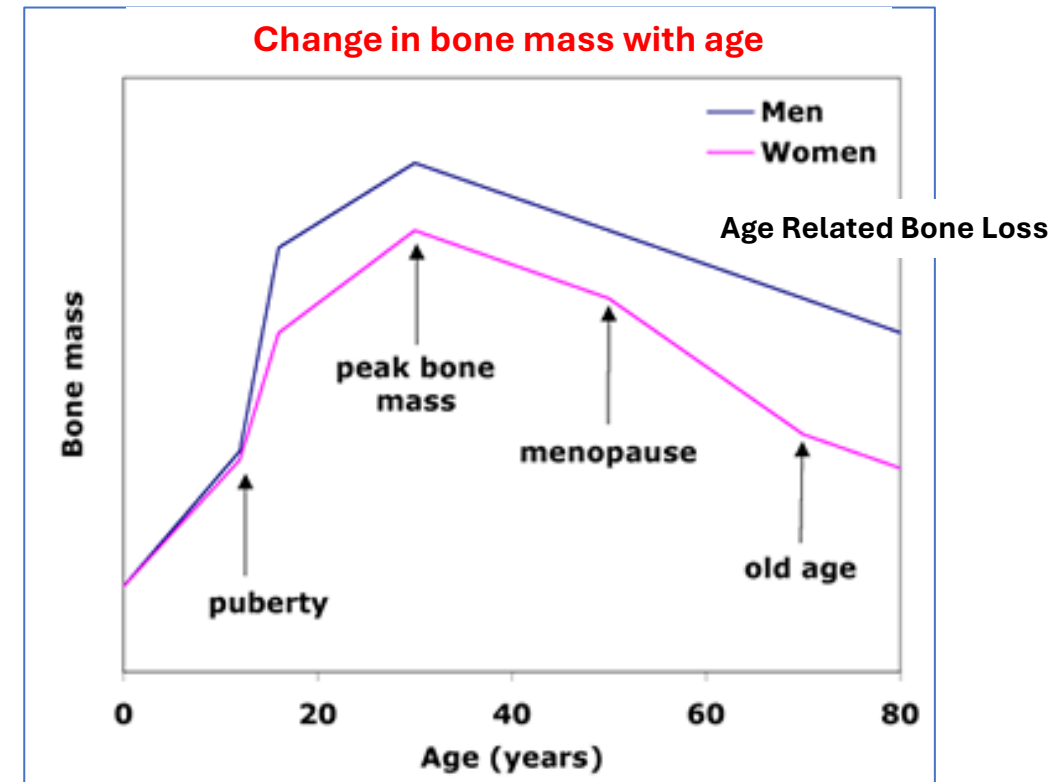
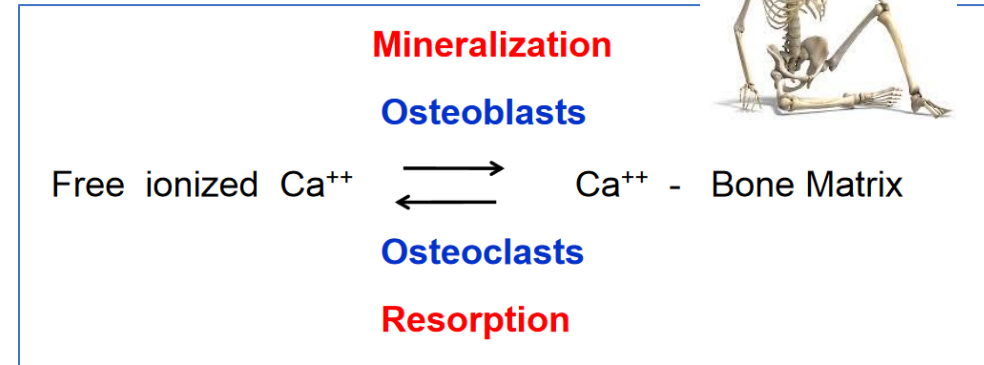
- Parathyroid hormone (PTH)
- Vitamin D (1,25-DHCC)

- **Hyperphosphatemia:**

- **Stimulates PTH** secretion → PTH decreases phosphate reabsorption in kidneys → Increased urinary excretion of phosphate
- Suppresses 1 α -hydroxylase in the kidneys (enzyme responsible for forming active form of vitamin D) → Reduced 1,25-dihydroxyvitamin D → decreased intestinal phosphate absorption

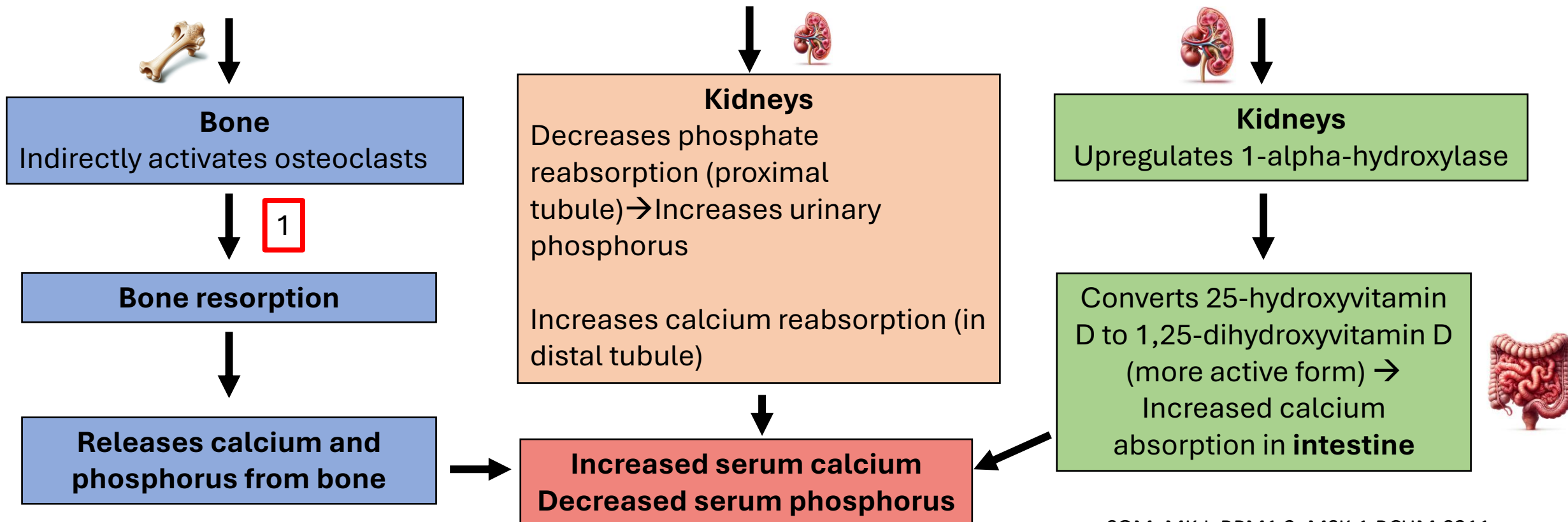
Bone Mass

- Bone tissue remodeled throughout life
 - **Osteoblasts build bone**
 - **Osteoclasts break down bone**
- Bone mass: Amount of osseous tissue in unit volume of bone
- Determinants of bone mass:
 - Intrinsic: genetics, gender, family history
 - Extrinsic:
 - Diet, body mass/habitus
 - Hormonal milieu (PTH, vitamin D, estrogen...)
 - Illnesses
 - Exercise, lifestyle



Parathyroid Hormone (PTH): Actions

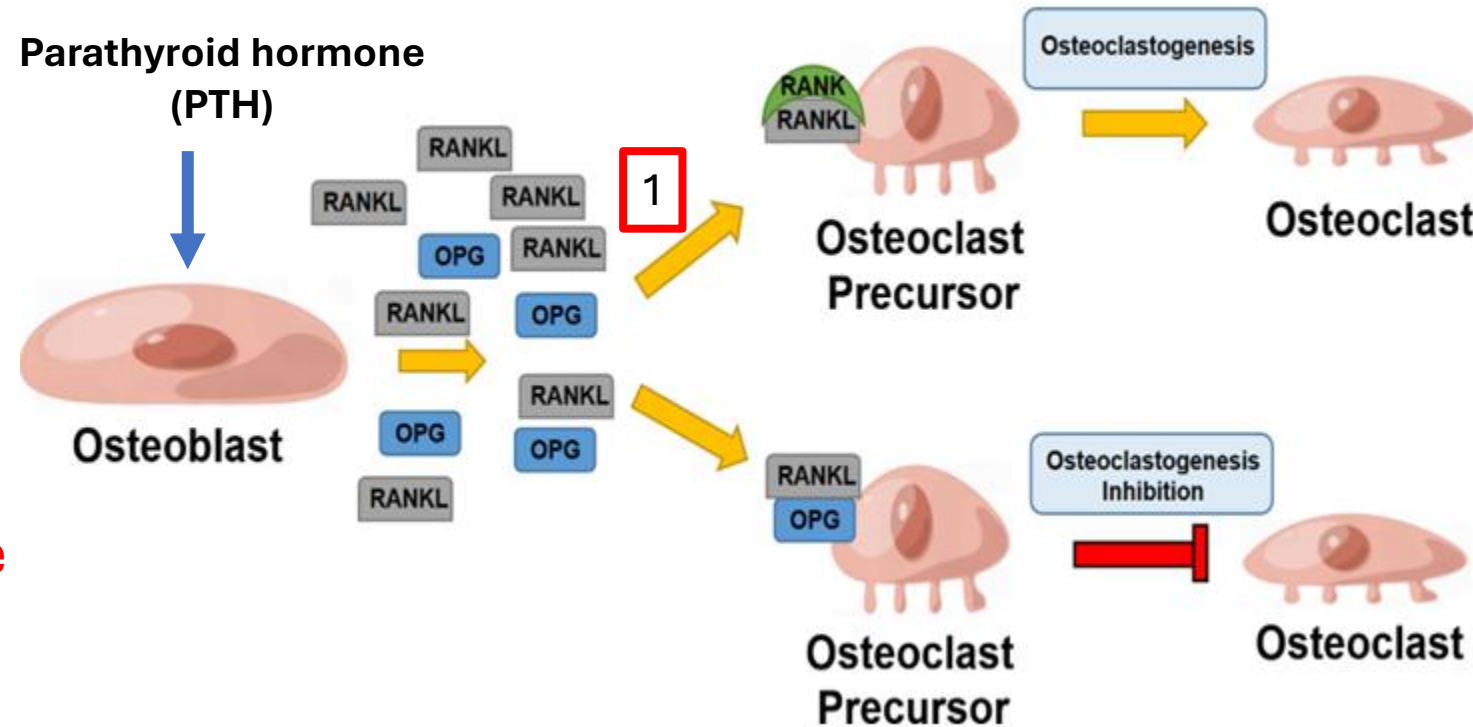
- Parathyroid hormone (PTH) is **released** in **response to hypocalcemia** and **hyperphosphatemia**
- PTH **increases serum calcium** and **lowers serum phosphorus**



PTH: Action on bone

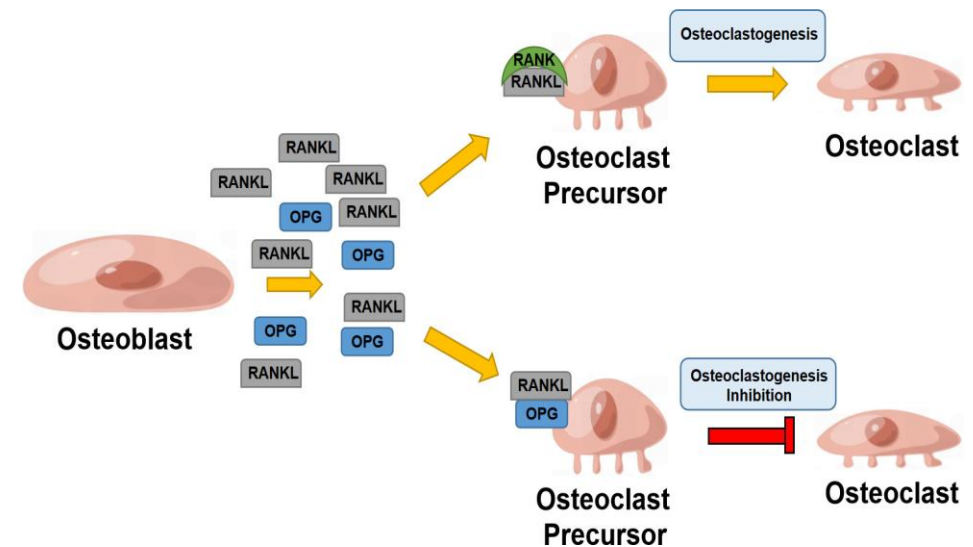
- Osteoblasts regulate osteoclast formation and differentiation
- **RANK** : Receptor on osteoclasts and its precursors
- **RANK-L**: Secreted by **osteoblasts** that **stimulates osteoclasts** by binding to RANK → **increases bone resorption**
- PTH stimulates osteoblasts to secrete **RANK-L** which **activates osteoclasts** → Bone **demineralization (resorption)**
 - PTH acts **indirectly** on osteoclasts

RANK: Receptor activator of nuclear factor κB
RANK-L: RANK-ligand



Bone Turnover: Role of OPG

- Osteoblasts also secrete **osteoprotegerin (OPG)**, a decoy receptor, that blocks interaction of RANK-L with RANK → decreased osteoclast differentiation & survival → decreased bone resorption.
- Bone turnover regulated by ratio of **OPG to RANK-L**



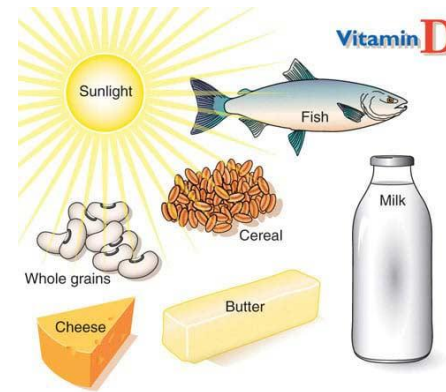
Regulation of bone turnover

- Parathyroid hormone (PTH), 1,25-dihydroxyvitamin D (1,25-DHCC) and gp130 cytokines (IL-6-type cytokines) stimulate RANK-L expression → enhancing bone resorption
 - In menopause → estrogen deficiency → secretion of gp130 cytokines (estrogen inhibits these cytokines) → increased osteoclastic activity → accelerated bone loss → increases **osteoporosis** susceptibility
 - Denosumab (monoclonal antibody drug) used in treatment of postmenopausal osteoporosis, acts similar to **OPG**.
- Wnt signaling stimulates **OPG** → suppressing bone resorption

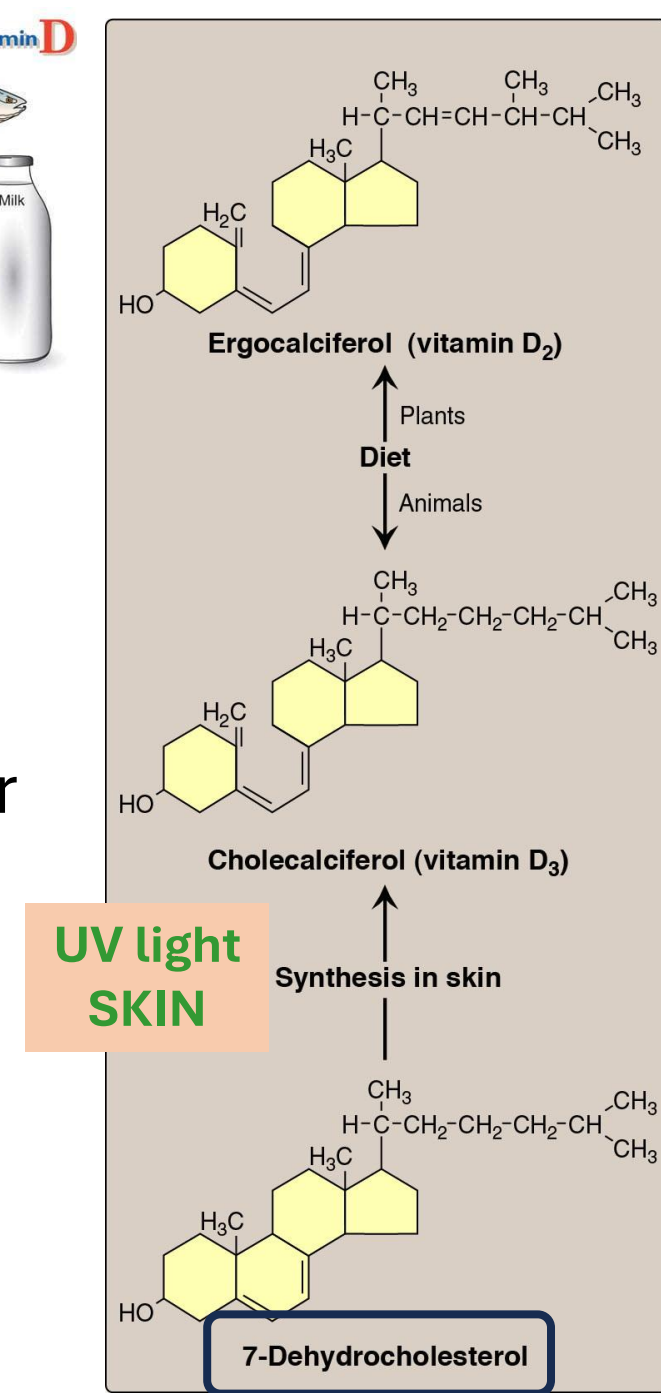
Summary: Effects of PTH on serum Calcium and Phosphorus Levels

- **PTH increases serum calcium** (Ca^{2+}) by:
 - ↑ Bone resorption (activates osteoclasts, via RANK-L by osteoblasts)
 - ↑ Ca^{2+} reabsorption in kidneys (Decreases urine calcium losses)
 - ↑ vitamin D activation in kidneys → ↑ Ca^{2+} absorption in the intestine
- **PTH decreases serum phosphorus levels:**
 - ↑ urinary excretion of phosphorus → Decreases serum phosphorus

Vitamin D (Cholecalciferol)

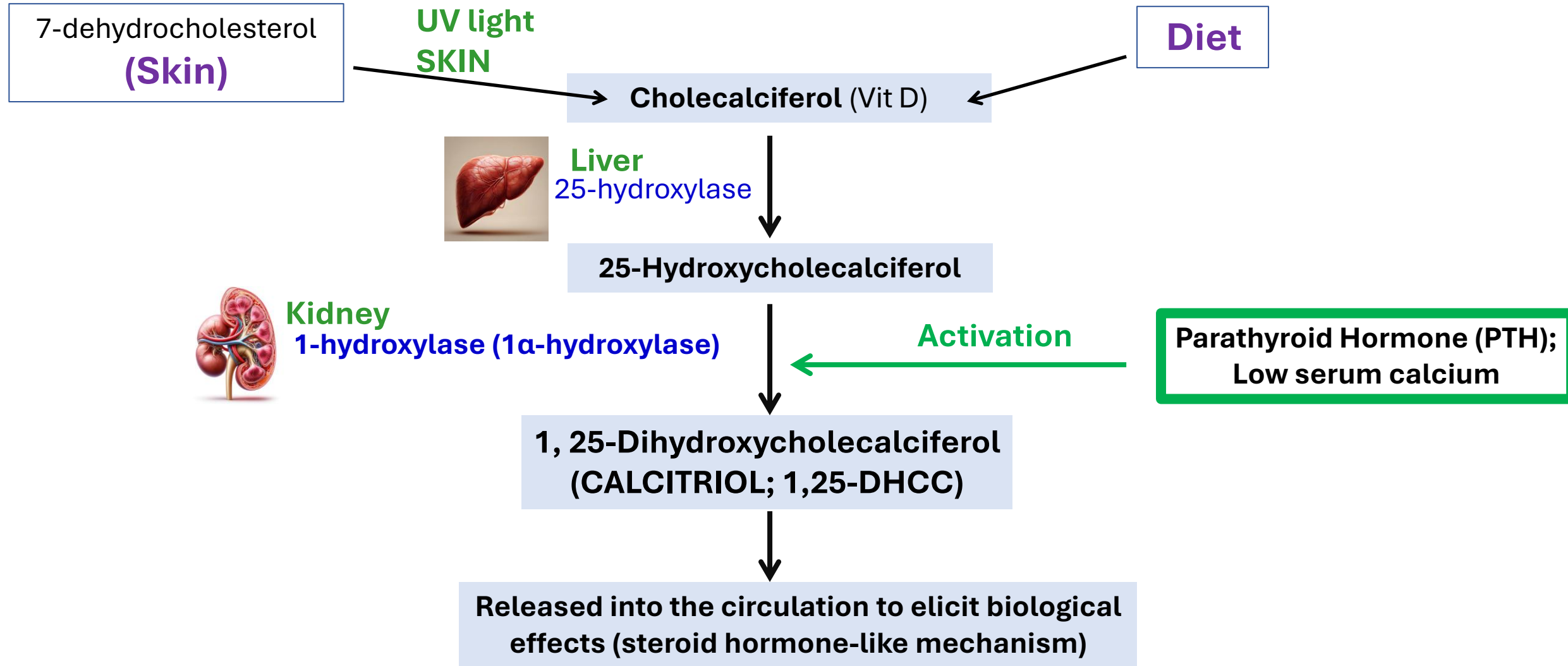


- Fat soluble vitamin
- Active form: 1,25-dihydroxycholecalciferol (Calcitriol)
 - **Steroid hormone-like action (intra-cellular receptors)**
- Ergocalciferol (vitamin D₂) of plant origin
- Cholecalciferol (vitamin D₃) of animal origin
- Sunlight converts **7-dehydrocholesterol** (precursor for cholecalciferol in skin) to **cholecalciferol (vitamin D)**
- **Formation of calcitriol:** Hydroxylation reactions
 - Liver (25-alpha hydroxylase)
 - **Kidney (1-alpha hydroxylase)**

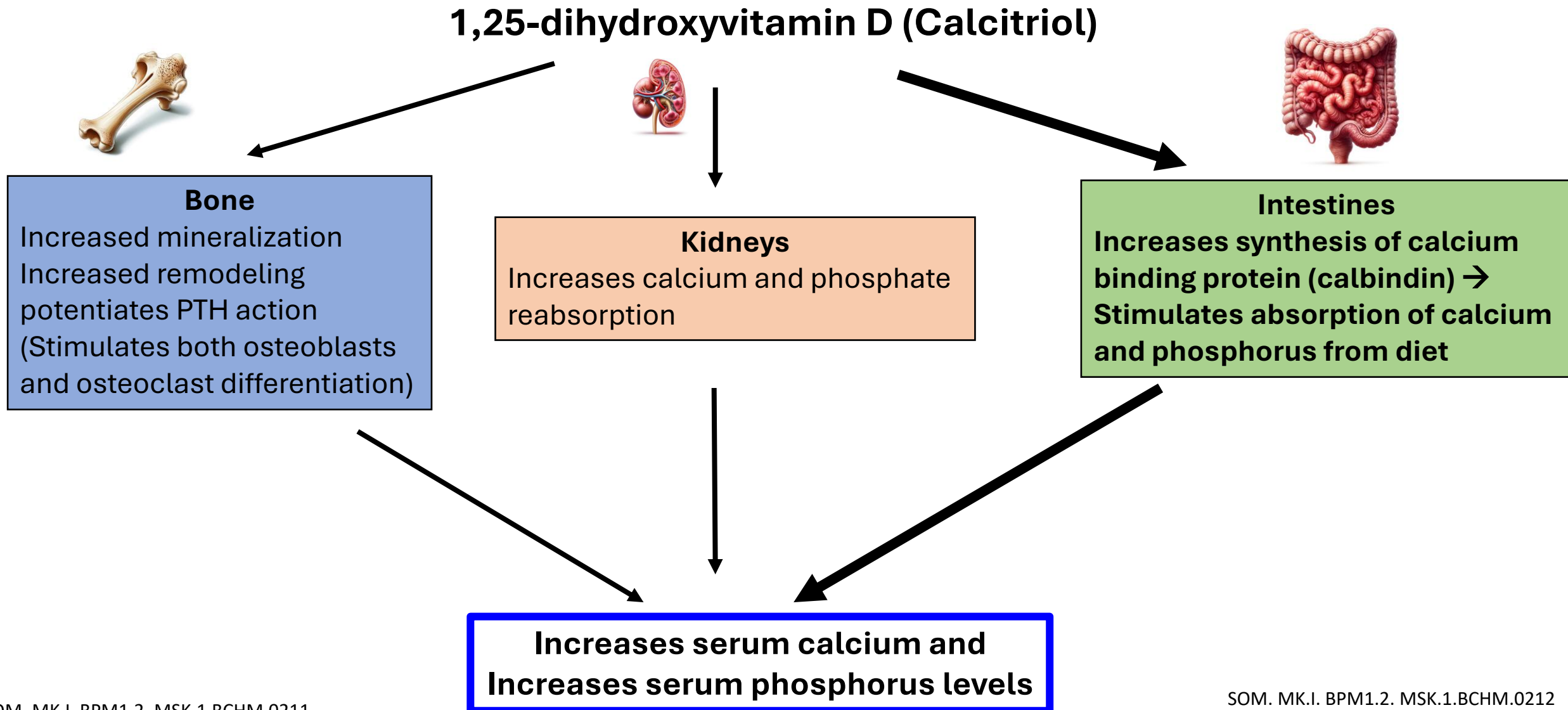


Formation of Active Vitamin D (Calcitriol): role of PTH

PTH activates kidney 1 α -hydroxylase $\rightarrow$ $\uparrow$ 1,25 Dihydroxycholecalciferol

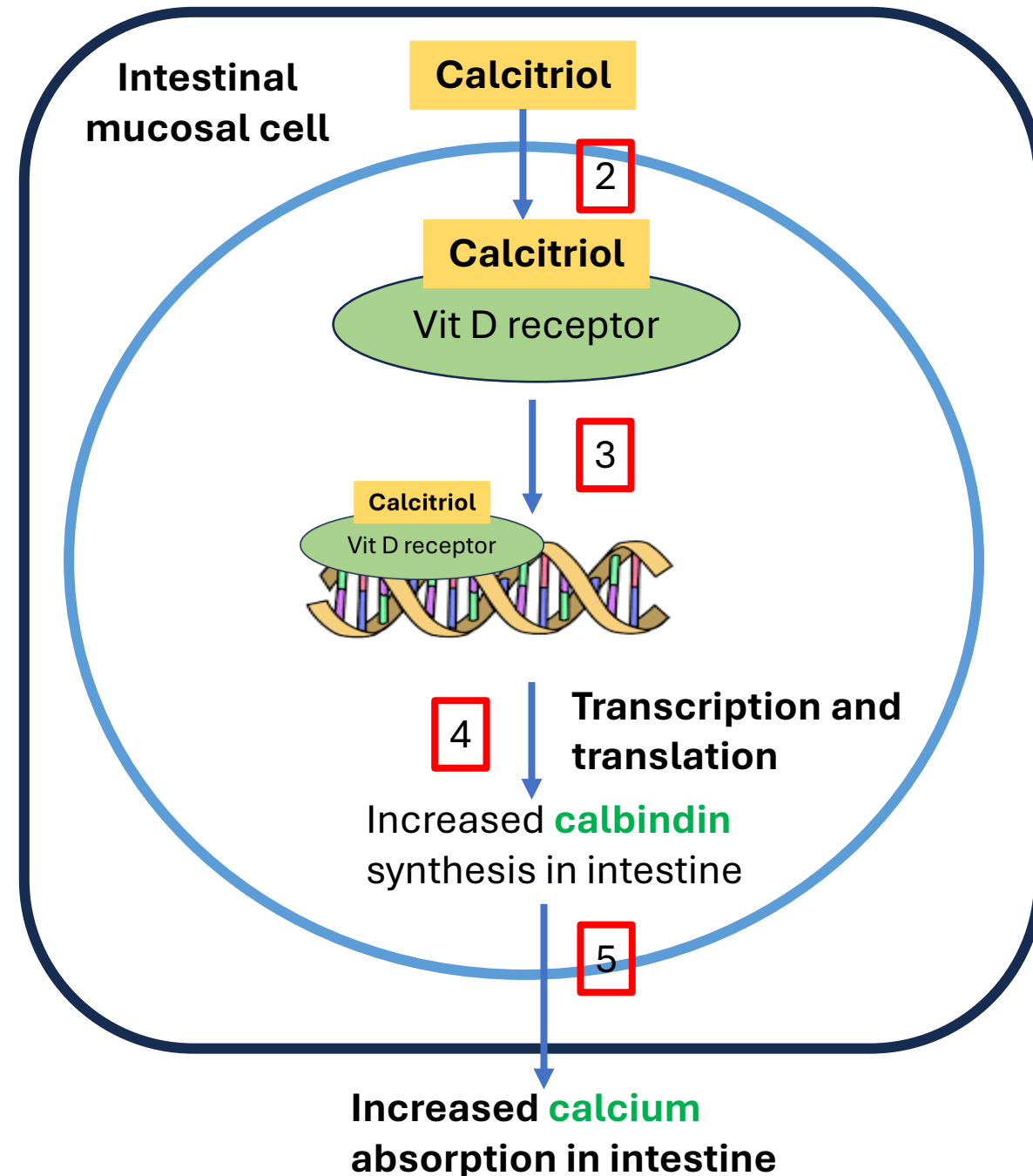


Calcitriol (1,25-dihydroxyvitamin D): Functions



Calcitriol mechanism of action: Intestine

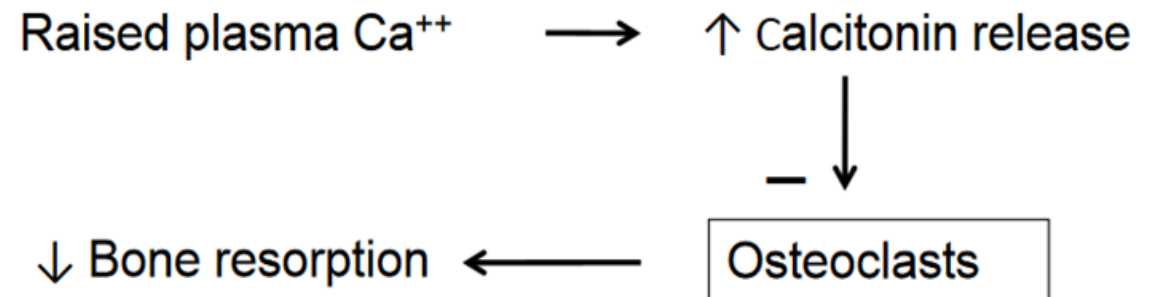
1. Calcitriol binds to intracellular receptors
2. Forms hormone-receptor complex
3. Hormone-receptor complex binds to DNA
4. Increased transcription and translation (Increased gene expression)
 - **Steroid hormone like action**
5. Increased synthesis of calcium binding proteins (**calbindins**) → **Increased dietary calcium absorption**
6. **Increases dietary phosphorus absorption**



Calcitonin

- Peptide hormone secreted by the thyroid C-cells
- Inhibits osteoclast activity → **Decreases bone resorption**
- **Reduces plasma calcium**, opposing the effects of PTH
- Important for bone mineralization in fetus and neonates
 - No significant function in adults
- **Do not confuse calcitriol (active vitamin D) and calcitonin**

Regulation of mineral metabolism and bone turnover.



Case 1

A 47-year-old woman comes to the emergency department because of **tingling sensation around her mouth** and **muscle cramps** in her legs. She reports that these symptoms started two days ago and have progressively worsened. Her medical history is significant for papillary thyroid carcinoma, for which **she underwent total thyroidectomy** three weeks ago.

On physical examination, she appears anxious. Her blood pressure is 136/84 mmHg, pulse is 98/min, and respirations are 20/min. **Light tapping anterior to the ear elicits twitching of the lower facial muscles.**

Laboratory studies show:

Serum calcium level: 7.2 mg/dL (normal: 8.6-10.2 mg/dL)

Serum phosphorus level: 5.1 mg/dL (normal: 2.5-4.5 mg/dL)

Parathyroid hormone (**PTH**) level: Markedly **low**.



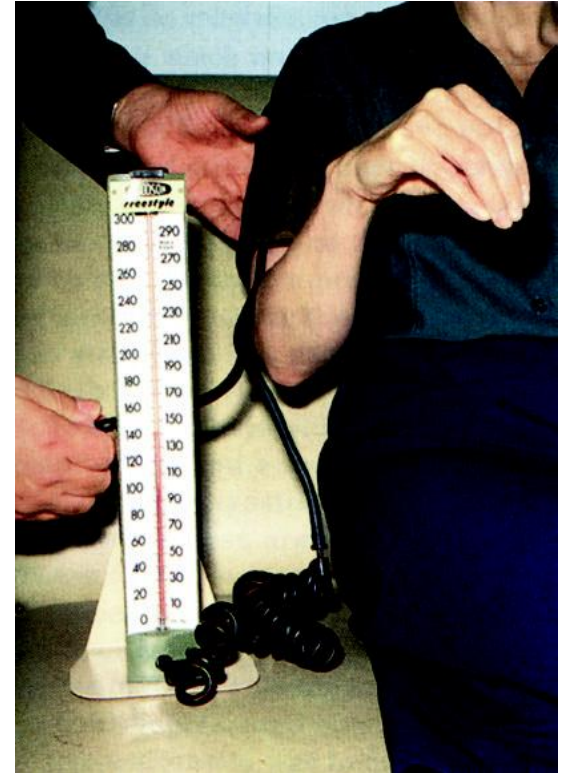
Trousseau's sign
(**Carpopedal spasm**)



Chvostek's sign
(**Twitching of facial muscles**)

Clinical Features of Hypocalcemia

- **Tetany**: increased neuromuscular excitability (hypocalcemia → Partial depolarization of nerves and muscles)
 - **Paresthesia**: tingling sensation in extremities and perioral area
 - **Spasms**: carpopedal spasm (thumb flexes into palm), laryngospasm (life-threatening)
 - Muscle cramps
 - **Chvostek** sign: twitching of facial muscles by tapping the facial nerve below and in front of the ear
 - **Trousseau** sign: Carpopedal spasm several minutes after blood pressure cuff inflation
 - Seizures
- Cardiac arrhythmias
- Chronic: Confusion, anxiety, psychosis and ophthalmologic complications



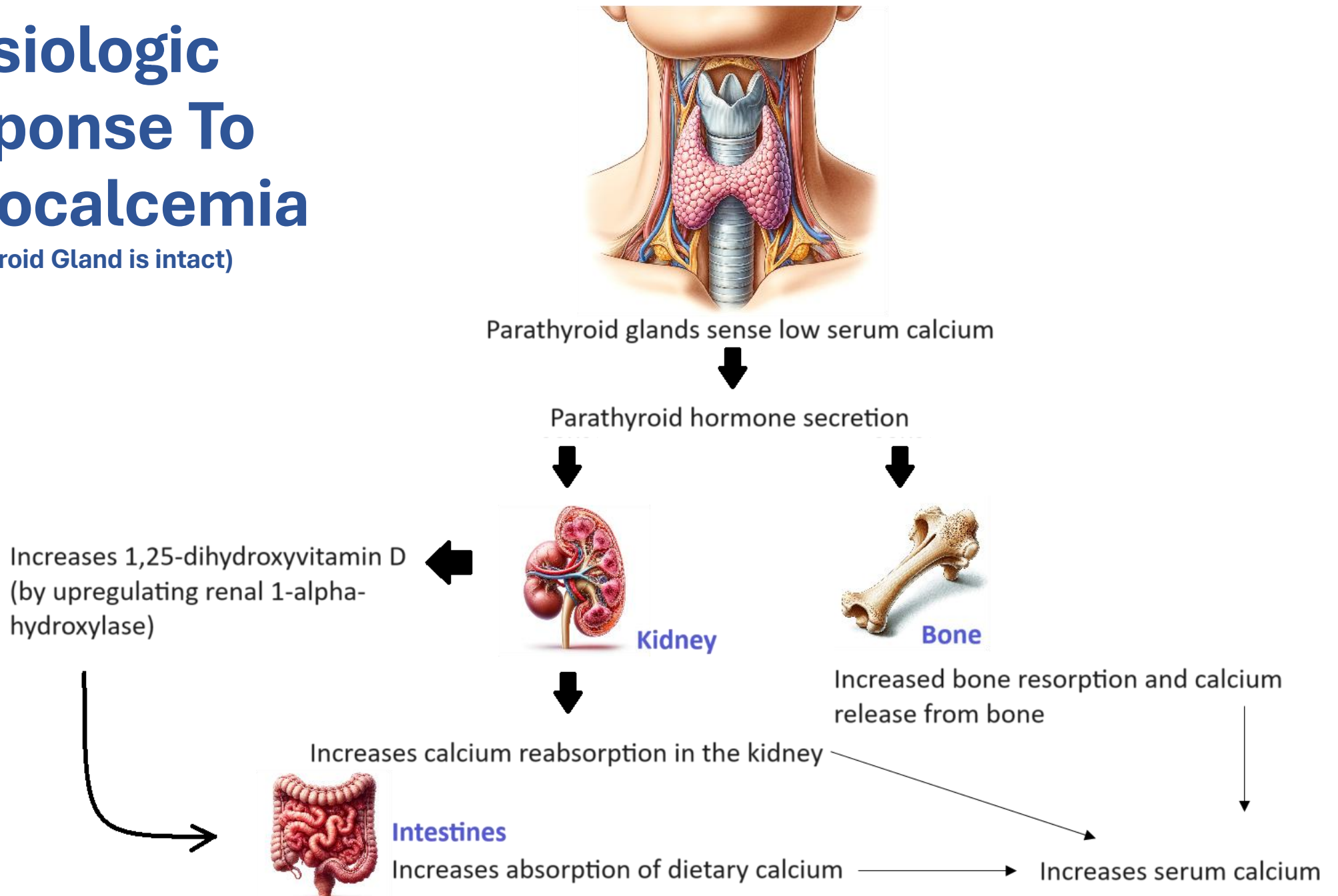
Carpal spasm (Trousseau sign; thumb adducts into the palm). (From *Rapid Review Pathology* 5th edition)

Causes of Hypocalcemia

- Hypocalcemia with **low PTH** (hypoparathyroidism):
 - Destruction of parathyroid glands (surgical, autoimmune)
 - Post-surgical: **after thyroidectomy** (proximity of parathyroid glands to thyroid gland)
 - Abnormal parathyroid gland development (e.g., **DiGeorge syndrome**)
 - **Hypomagnesemia** → impairs PTH release (functional hypoparathyroidism)
- Hypocalcemia with **high PTH** (Secondary hyperparathyroidism):
 - **Vitamin D deficiency**
 - **Chronic kidney disease**

Physiologic Response To Hypocalcemia

(If Parathyroid Gland is intact)



Laboratory Findings of Hypocalcemia

- Low serum calcium levels
- 25 hydroxyvitamin D levels: Low if nutritional deficiency of vitamin D;
In Vitamin D deficiency, PTH levels are elevated (secondary)
- 1,25 dihydroxyvitamin D levels: Low if renal insufficiency
- PTH levels: Low after parathyroidectomy

Laboratory Findings of Hypocalcemia

Cause	PTH	Calcium	Phosphorus	25(OH) D	1,25(OH) ₂ D
Hypoparathyroidism	Low	Low	Elevated	Normal	Normal or Low
Vitamin D Deficiency	Elevated	Low	Low or Low Normal	Low	Low Normal or Low
Chronic Kidney Disease	Elevated	Low	Elevated	Normal or Low*	Low

Case 2

A 3-year-old boy is brought to the pediatrician by his parents due to concerns about his growth and development. His parents noticed that his legs appear to be **bowing outward**.

The family lives in a northern city and his mother admits that they rarely go outside during the winter months.

The child's diet is primarily vegetarian, and he drinks almond milk instead of cow's milk. Examination reveals a height and weight below the 10th percentile for his age, **widened wrists and ankles**, and a **prominent forehead**.

Laboratory tests show:

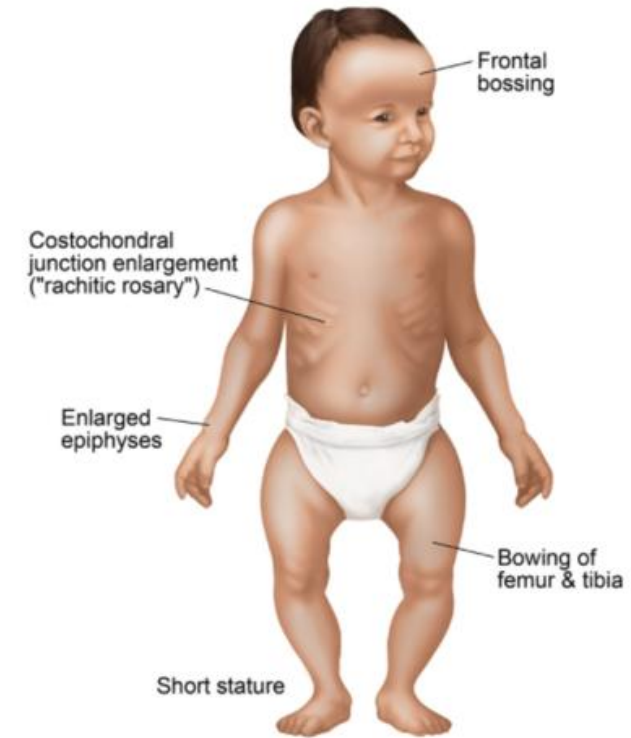
Serum calcium: 8.2 mg/dL (normal: 8.6-10.2 mg/dL)

Serum phosphorus level: 2.3 mg/dL (normal: 2.5-4.5 mg/dL)

Alkaline phosphatase (ALP): 800 U/L (25-100 U/L)

Low 25-hydroxyvitamin D levels

Clinical manifestations of rickets



Enlarged epiphysis



Bowing of legs



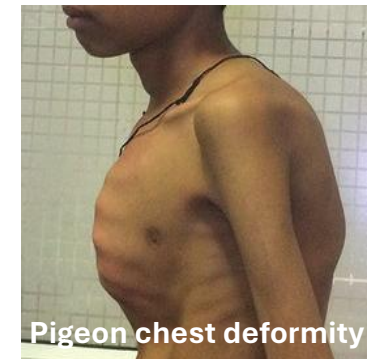
bow-leg deformity

Causes of Vitamin D deficiency

- **Inadequate exposure to sunlight** (decreased photoconversion of 7-dehydrocholesterol to cholecalciferol)
 - Winter; excess melanin
- **Renal failure** (most common cause in adults)
 - Decreased **1- α hydroxylation** of vitamin D
- **Inadequate dietary intake**
 - Vegan diets, lack of vitamin-D fortified food
- **Fat malabsorption** (micelle formation required for vitamin D absorption)
 - Inflammatory bowel disease, cystic fibrosis
- **Chronic liver disease** (decreased 25-hydroxyvitamin D synthesis)
- **Exclusive breastfeeding without vitamin D supplements** (breast milk low in vitamin D)

Clinical Features of Vitamin D deficiency

- Both adults and children:
 - Pathologic fractures (excess of unmineralized osteoid)
 - **Tibial bowing** (soft bones)
 - Tetany (low serum ionized calcium)
- **Rickets** seen in **children** only: (before fusion of epiphyses)
 - **Rachitic rosary** (defective mineralization, overgrowth of unmineralized cartilage in costochondral cartilage)
 - **Craniotabes** (soft skull bones, delayed suture and fontanelle closure)
 - Frontal bone thickening, forehead bossing
 - Short stature
 - Pectus carinatum (pigeon breast)
- **Osteomalacia** (soft bone) in **adults** due to defective remodeling:
 - Weakening of bones, frequent fractures



Laboratory Findings of Vitamin D deficiency

- **Decreased serum calcium** (total and ionized)
 - Low vitamin D → Decreased calcitriol formation → Decreased calcium absorption in intestine → Low serum calcium level
- **Decreased serum phosphorus**
 - Poor intestinal phosphorous absorption and hyperparathyroidism
- **Elevated PTH (secondary hyperparathyroidism)**
 - Low vitamin D → Decreased calcitriol formation → Decreased calcium absorption in intestine → Low serum calcium level → Stimulates PTH secretion (**secondary hyperparathyroidism**)
- **Elevated alkaline phosphatase (ALP)**

Radiologic and lab findings

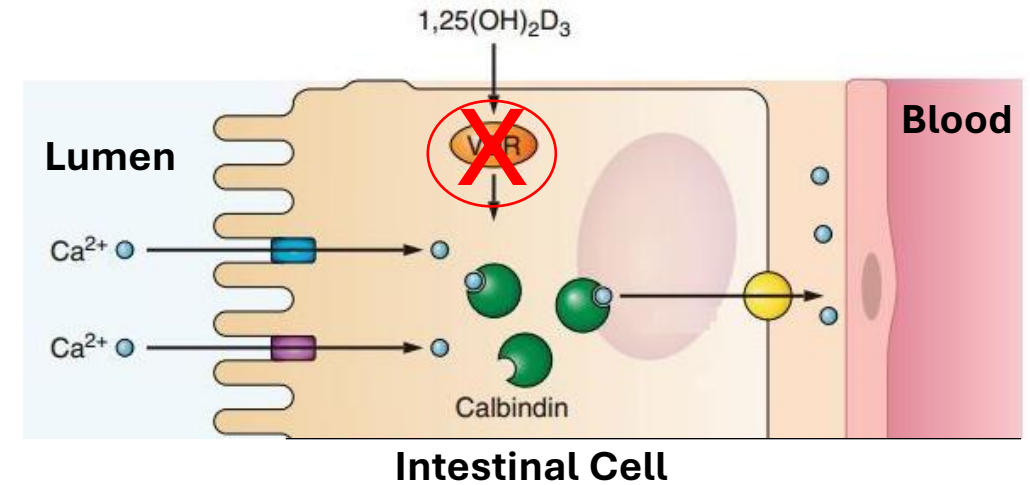
Review from Histology



(A) **Rickets**. Anteroposterior radiograph of the wrist and hand in a 3-year-old child with nutritional rickets. The child had been put on a strict diet without dairy products. Note the **widening, cupping, and fraying of the distal radius** (arrowhead) and ulna metaphyses with an associated increase in the thickness of the growth plate (arrow). These changes are the consequence of disordered endochondral growth.

(B) Normal. Radiograph of the hand of a healthy 3-year-old child, without rickets.

Hereditary Resistance to Vitamin D



- Autosomal recessive disorder
- Mutations in gene encoding **vitamin D receptor** → reduced binding of calcitriol → **decreased calcium and phosphate absorption in intestine** → Low serum calcium and phosphorus levels
- Clinical features similar to rickets + alopecia (hair loss)
- Elevated plasma 1,25-dihydroxyvitamin D levels
 - Low serum calcium level stimulates calcitriol synthesis

Case 3

- A 45-year-old man comes to the physician because of a 1-year history of recurrent abdominal pain.
- He has had a history of renal stones.
- Laboratory studies show:
 - Serum calcium level: 11.5 mg/dL (normal: 8.6-10.2 mg/dL)
 - Serum phosphorus level: 2 mg/dL (normal: 2.5-4.5 mg/dL)
 - Serum PTH levels: Elevated

Diagnosis: Primary hyperparathyroidism

Clinical Features of Hypercalcemia

- Nephrolithiasis (calcium oxalate kidney stones)
- Bone pain, arthralgias, myalgias, fractures
 - Excessive PTH → Bone resorption by stimulation of osteoclasts
- Constipation, abdominal pain, nausea, vomiting
 - Due to decreased excitability in gastrointestinal smooth muscle cells
- Peptic ulcer disease
 - Due to calcium stimulating gastrin → increased gastric acidity
- Pancreatitis
 - Calcium deposition in pancreatic duct → trypsinogen activation within pancreatic parenchyma → inflammation within the pancreas
- Neuropsychiatric: Anxiety, depression, fatigue, cognitive dysfunction

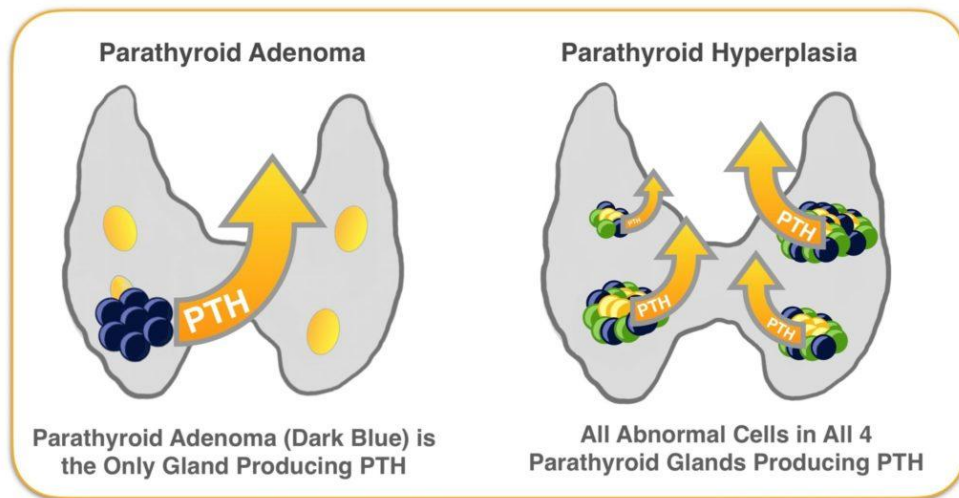
Causes of Hypercalcemia: Differentiate

Primary hyperparathyroidism

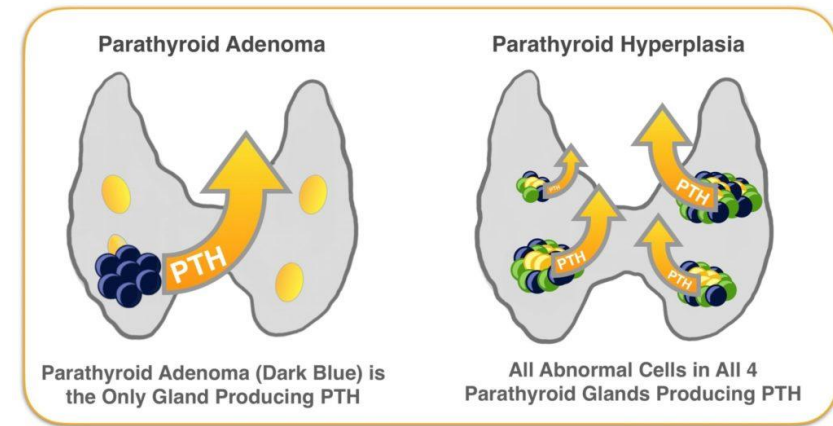
- **Over secretion of PTH**
- **Serum calcium: Elevated**
- **PTH levels: Elevated**
- **Serum phosphorus: Low**

Vitamin D intoxication (Hypervitaminosis D)

- Vitamin D over-supplementation
- **Serum calcium: Elevated**
- **PTH levels: Low**
- **Serum phosphorus: Elevated**



Primary Hyperparathyroidism



- **Excessive PTH secretion** (PTH levels elevated)
- **Elevated serum calcium levels** because PTH
 - Stimulates bone resorption (osteoclasts) → Bone pain
 - Increases calcium reabsorption in kidneys → Low urinary calcium excretion
 - Increases renal 1-hydroxylase activity → Increases calcitriol formation → Increases intestinal calcium absorption
- **Low serum phosphorus levels** because
 - PTH increases phosphorus excretion in urine

Hypervitaminosis D (vitamin D toxicity)

- Vitamin D is stored and slowly metabolized (All fat-soluble vitamins)
- High doses (100,000 IU for weeks/months) can cause loss of appetite, nausea, thirst, and stupor (non-specific symptoms)
- **Elevated serum calcium and phosphate levels**
- Enhanced intestinal calcium and phosphorus absorption and bone resorption → hypercalcemia → calcium deposition in different tissues (e.g., arteries and kidneys) → **Soft tissue calcification**

Hypophosphatemia

- Causes:
 - Insufficient intestinal absorption (e.g., impaired dietary intake, malabsorption, vitamin D deficiency)
 - Increased renal phosphate excretion (e.g., hyperparathyroidism)
- Clinical features:
 - Arrhythmias
 - Muscle weakness, respiratory muscle dysfunction (lack of ATP)
 - Osteomalacia (phosphorus needed to mineralize bone)
 - RBC hemolysis (lack of ATP)

Hyperphosphatemia

- Causes:
 - Decreased renal excretion (e.g., renal failure)
 - Increased intestinal absorption (Vitamin D toxicity)
 - Hypoparathyroidism (Low PTH)
- Clinical features:
 - Often asymptomatic
 - Calcification in soft tissues, skin and organs
 - Nephrolithiasis (kidney stones)

Condition	Serum PTH Levels	Serum Calcitriol Levels	Serum Calcium Levels	Serum phosphorus levels
Hypoparathyroidism (<i>Destruction of gland</i>)	↓	↓ (Low PTH levels result in low 1-hydroxylase activity)	↓	↑
Vitamin D deficiency Rickets- Children Osteomalacia - Adults	↑ (Low serum calcium stimulates PTH secretion: Secondary hyperparathyroidism)	↓ (Less sunlight exposure; renal failure → low 1-hydroxylase activity; low vit D in diet; liver disease)	↓ (Poor absorption of calcium)	↓ (Poor absorption of phosphorus and PTH)
Hyperparathyroidism (Primary) (<i>Intrinsic problem with gland</i>)	↑ (Increased PTH secretion from the parathyroid gland or elsewhere)	↑ (Excessive activity of 1-hydroxylase due to elevated PTH levels)	↑ (Increased bone resorption; increased tubular calcium reabsorption; Increased dietary calcium absorption)	↓ (Low serum phosphorus due to loss of phosphate in urine)
Vitamin D toxicity (Hypervitaminosis D)	↓ (Increased serum calcium → low PTH secretion)	↑ (Elevated 25-hydroxy vit D levels due to excessive intake)	↑ (Elevated calcitriol → Increased dietary calcium absorption)	↑ (Elevated calcitriol → Increased dietary phosphorus absorption)